ECE 340: Semiconductor Devices

Lecture 19: P-N Junction

Prof. Hyunseok Kim

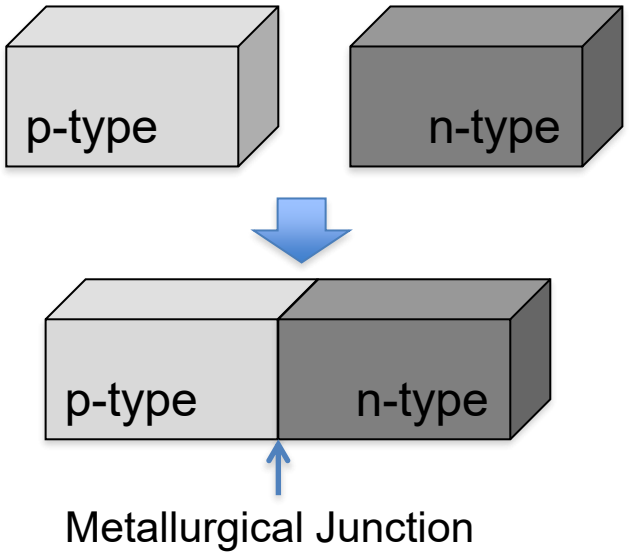
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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Definitions

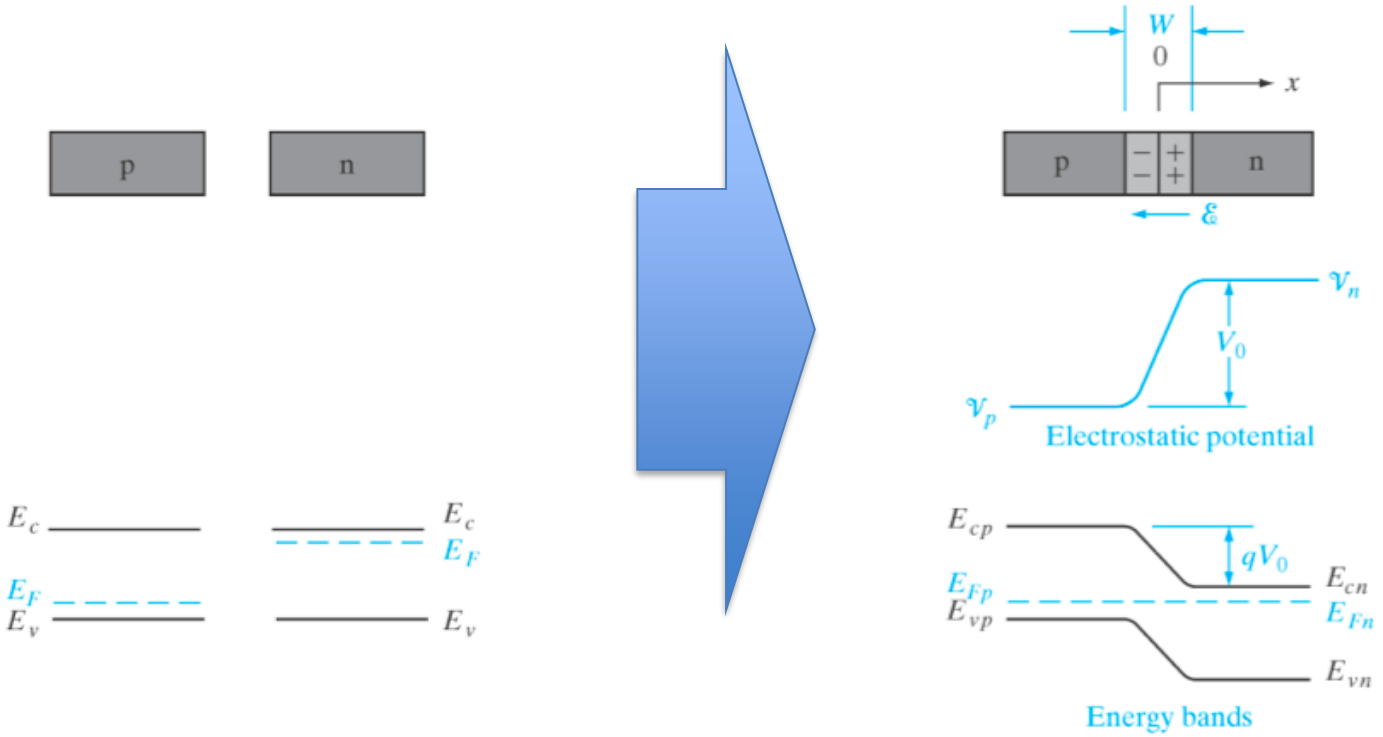
- ***Metallurgical Junction:*** The specific point at which the transition between p-type and n-type material occurs



- ***Step Junction:*** A junction in which there is an **abrupt transition** in the material properties (doping concentration & type) between each side, with each side uniform
- ***Graded Junction:*** A junction in which the material properties (compensated doping concentration & type) change in the region of the junction

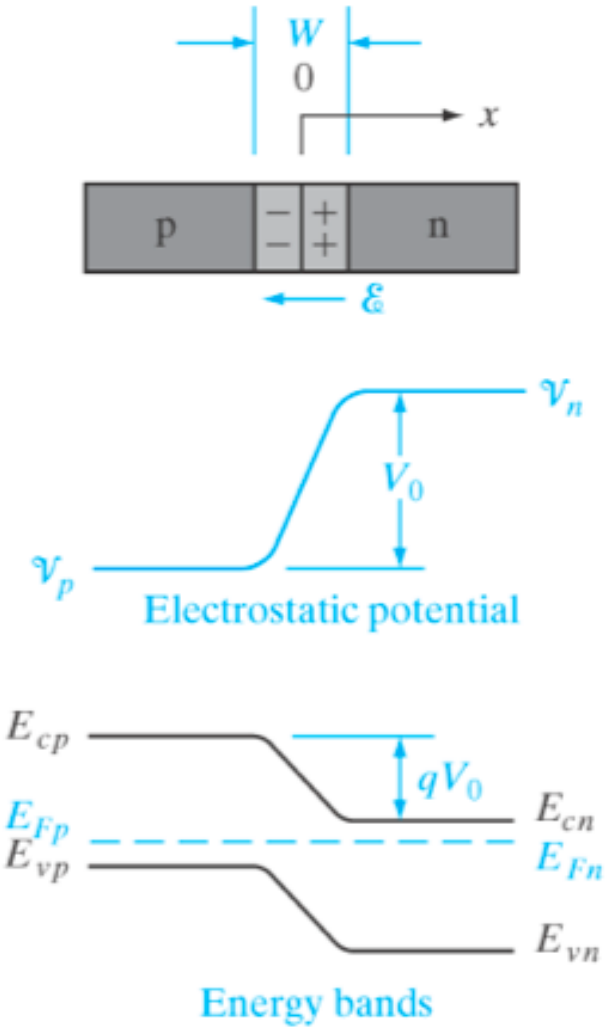
Equilibrium Condition

Equilibrium pn Junction



Equilibrium pn Junction

- In equilibrium, there is no net current flow across the junction, so:
  - J_p(Drift)+J_p(Diffusion)=0
  - J_n(Drift)+J_n(Diffusion)=0
- The electric field balances the diffusion current
- The potential difference $V_o=V_n-V_p$ develops in the direction opposite to the electric field E
- V_o is the "***contact potential***" which is a ***built-in potential*** barrier
  - Contact potential cannot be measured
- W is the "***transition region***"
  - Also called "depletion region" and "space charge region"
- We can calculate V_o from the separation in the Fermi levels (this will be shown later)



Contact Potential
(Built-in Potential)

Derivation of Contact Potential

In Equilibrium, there is no net current flow:

$$J_p(x) = q\mu_p p(x)\mathcal{E}(x) - qD_p \frac{dp(x)}{dx} = 0$$

so $$\frac{\mu_p}{D_p}\mathcal{E}(x) = \frac{1}{p(x)}\frac{dp(x)}{dx}$$

but since $$\frac{D}{\mu} = \frac{kT}{q}$$ and $$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx}$$

we have $$-\frac{q}{kT}\frac{d\mathcal{V}(x)}{dx} = \frac{1}{p(x)}\frac{dp(x)}{dx}$$

Potential on p-side

Hole concentration on p-side

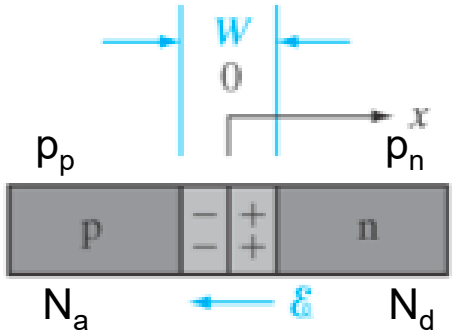
This is integrated over the limits $\mathcal{V}_p$ to $\mathcal{V}_n$ and p_p to p_n

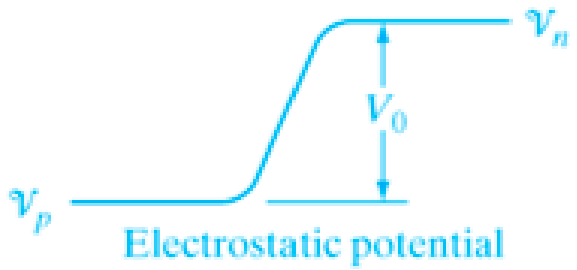
across the space charge region:

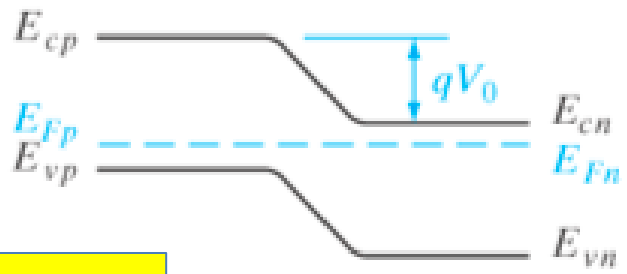
$$-\frac{q}{kT}\int_{\mathcal{V}_p}^{\mathcal{V}_n} d\mathcal{V} = \int_{p_p}^{p_n}\frac{1}{p}dp \text{ so } -\frac{q}{kT}\left(\mathcal{V}_n - \mathcal{V}_p\right) = \ln p_n - \ln p_p = \ln\frac{p_n}{p_p}$$

$$\text{so } \mathcal{V}_n - \mathcal{V}_p = V_o = -\frac{kT}{q}\ln\frac{p_n}{p_p} = \frac{kT}{q}\ln\frac{p_p}{p_n} = \frac{kT}{q}\ln\frac{N_a}{\left(n_i^2/N_d\right)} = \frac{kT}{q}\ln\frac{N_a N_d}{n_i^2}$$

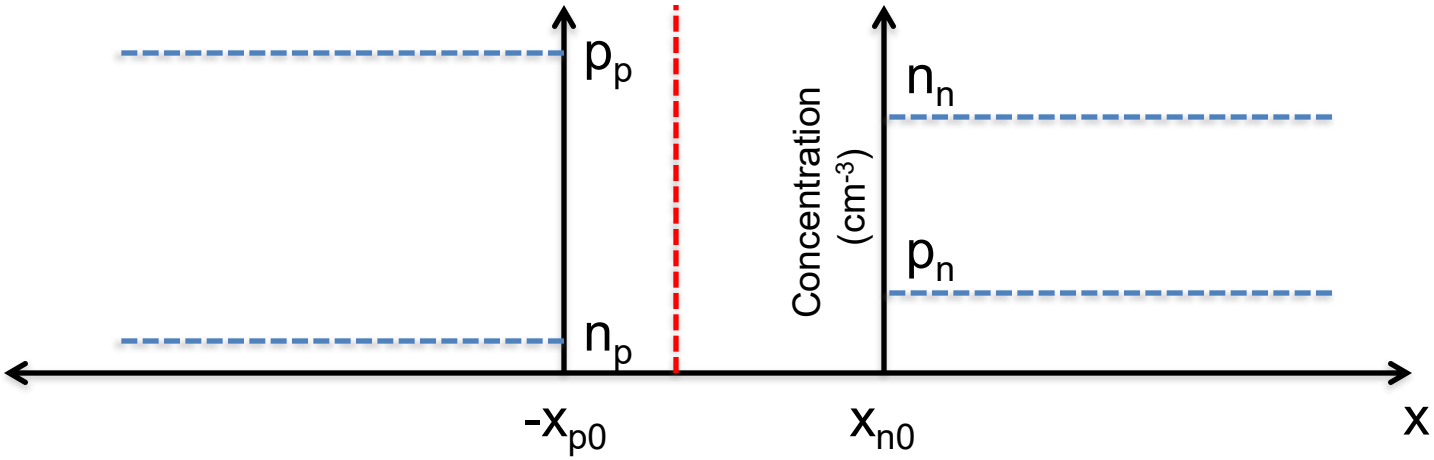
Since $$V_o = \frac{kT}{q}\ln\frac{p_p}{p_n}$$ and $p_p n_p = n_i^2 = p_n n_n$, $$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$$







Carrier Concentrations



$$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$$

$$p_p = p_n e^{qV_o/kT} \quad \text{or} \quad p_n = p_p e^{-qV_o/kT}$$

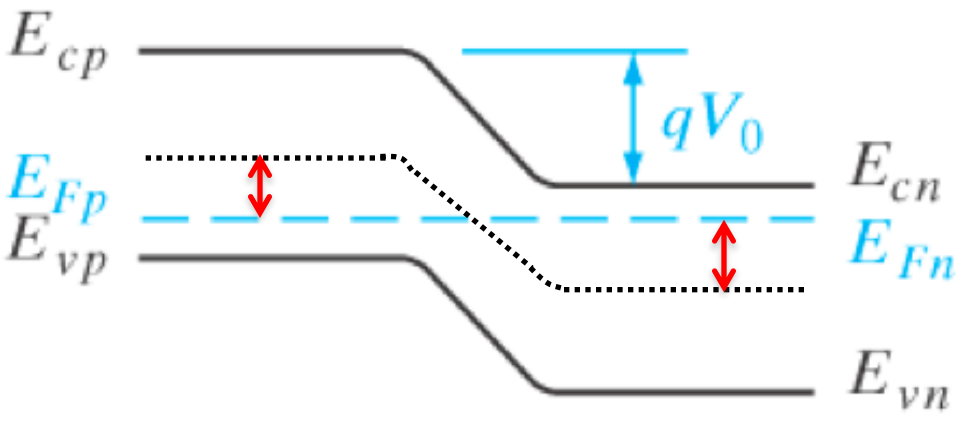
$$n_p = n_n e^{-qV_o/kT} \quad \text{or} \quad n_n = n_p e^{qV_o/kT}$$

Calculation of Contact Potential

Method 1:

- Use the relationship we derived last time

$$V_0 = \frac{kT}{q}\ln\frac{p_p}{p_n} = \frac{kT}{q}\ln\frac{N_a N_d}{n_i^2}$$



Method 2:

- Calculate the separation between the Fermi level and the intrinsic level on both sides of the junction, and sum the results

$$E_{ip} - E_F = kT\ln\frac{p_p}{n_i}$$

$$E_F - E_{in} = kT\ln\frac{n_n}{n_i}$$

$$V_0 = \left(E_{ip} - E_F\right) + \left(E_F - E_{in}\right)$$

$$= kT\ln\frac{p_p}{n_i} + kT\ln\frac{n_n}{n_i}$$

Example: Contact Potential Calculation

Method 2 :

$$E_{fn} - E_{in} = kT \ln \frac{n_{no}}{n_i}$$

$$= 0.026eV \left(\ln \frac{10^{16}}{1.5x10^{10}} \right)$$

$$E_{fn} - E_{in} = 0.347eV$$

$$E_{ip} - E_{fp} = kT \ln \frac{p_{po}}{n_i}$$

$$= 0.026eV \ln \frac{10^{17}}{1.5x10^{10}}$$

$$E_{ip} - E_{fp} = 0.407eV$$

$$qV_0 = 0.407 + 0.347eV = 0.754eV$$

$$V_0 = 0.754V$$

$p_{po} \cong N_a^- = 10^{17} cm^{-3}$

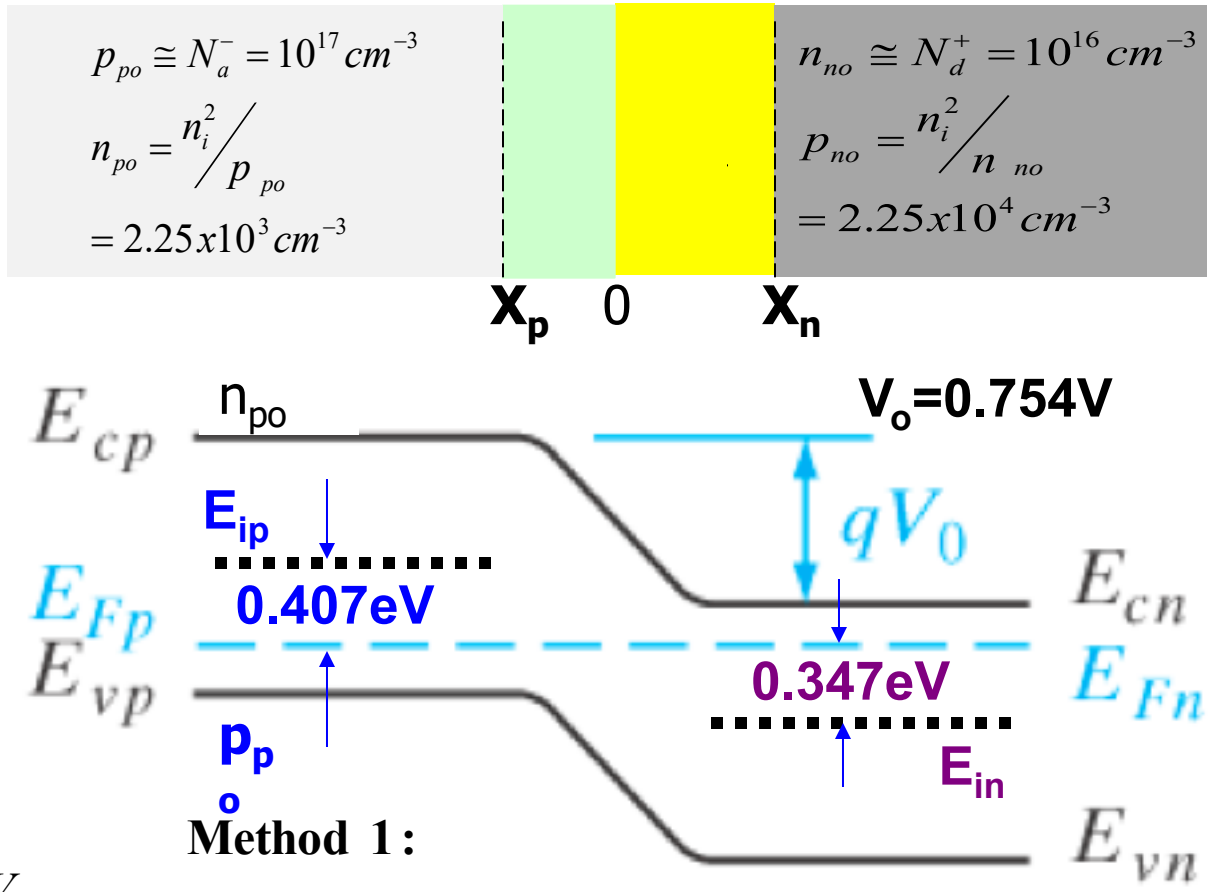
$n_{po} = n_i^2 / p_{po}$

$= 2.25x10^3 cm^{-3}$

$n_{no} \cong N_d^+ = 10^{16} cm^{-3}$

$p_{no} = n_i^2 / n_{no}$

$= 2.25x10^4 cm^{-3}$



Method 1 :

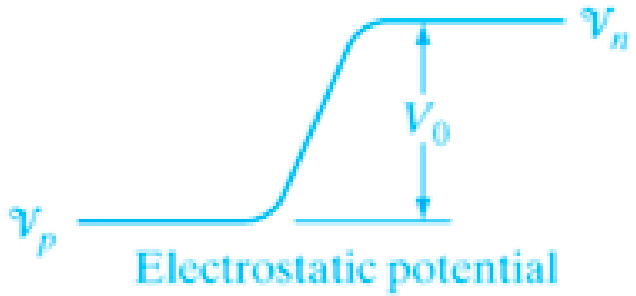
$$V_o = \frac{kT}{q} \ln \frac{p_{po}}{p_{no}} = \frac{kT}{q} \ln \frac{n_{no}}{n_{po}}$$

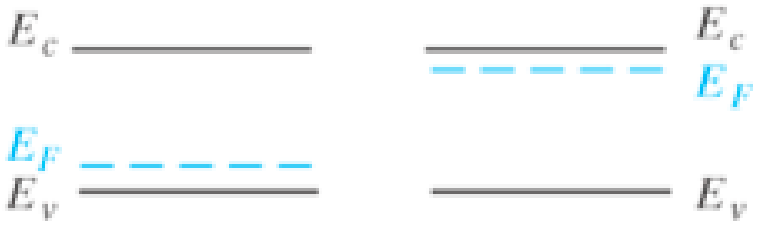
$$V_o = 0.026V \left(\ln \frac{10^{17}}{2.25x10^4} \right) = 0.754V$$

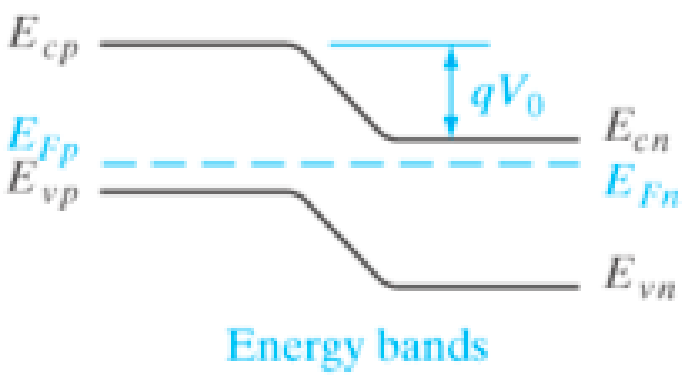
Equilibrium Fermi Levels

Comments

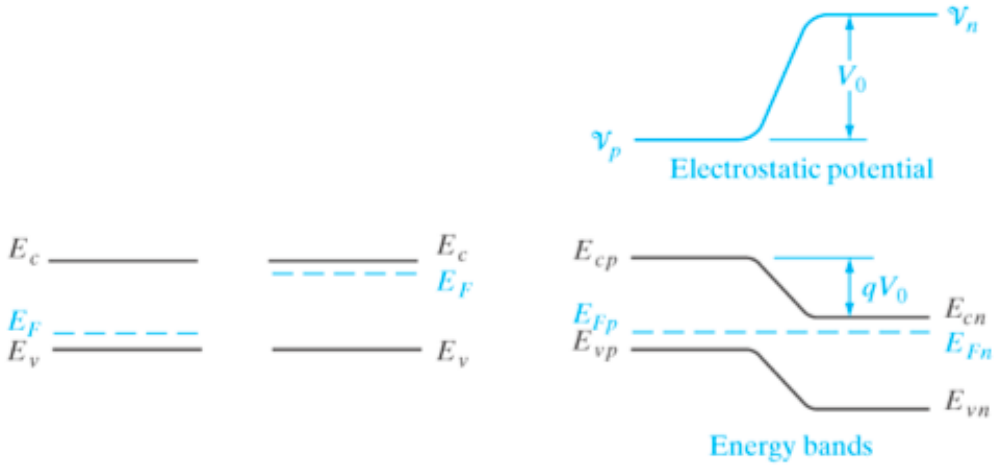
- In equilibrium, the Fermi level is flat
- The equilibrium energy band separation is “q” times the contact potential
- The contact potential is a built-in potential barrier – don’t think of this as an external voltage that appears across the device







Band Separation and Contact Potential



Since p_n and p_p are given by their equilibrium values away from the depletion (transition) region:

$$e^{qV_o/kT} = \frac{p_p}{p_n} = \frac{N_v e^{-(E_{Fp}-E_{vp})/kT}}{N_v e^{-(E_{Fn}-E_{vn})/kT}} \text{ so:}$$

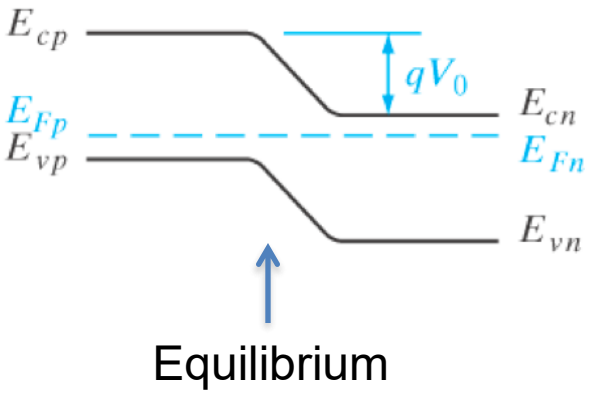
$$e^{qV_o/kT} = e^{(E_{Fn}-E_{vn})/kT} e^{(E_{vp}-E_{Fp})/kT} \text{ and, since } E_{Fn} = E_{Fp}:$$

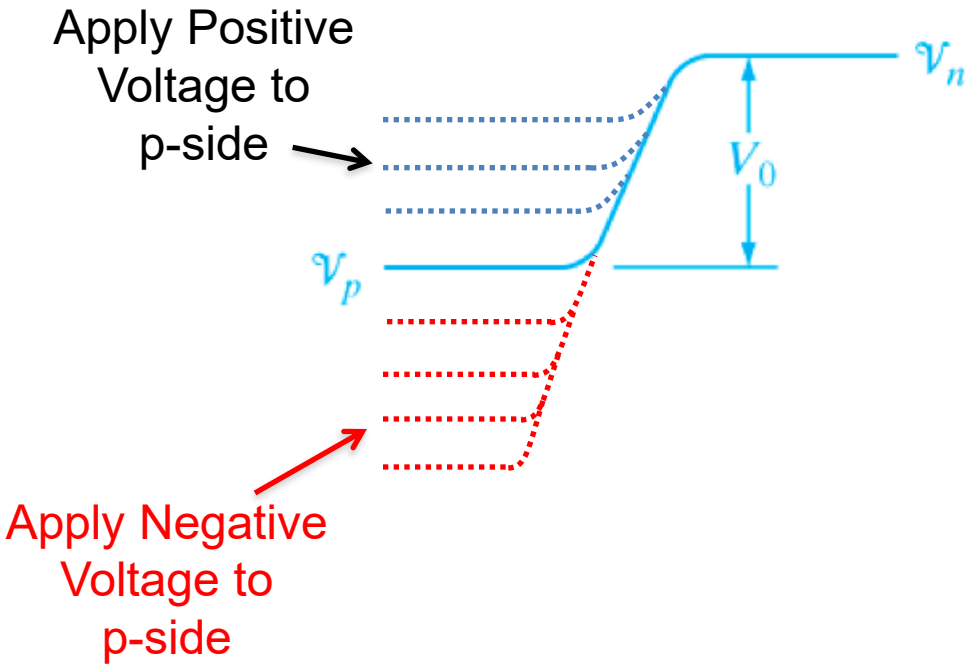
$$qV_o = E_{vp} - E_{vn}$$

The band separation is "q" times the contact potential.

- Applying an external bias raises or lowers the potential barrier created by the contact potential!

Externally applied potential: a preview



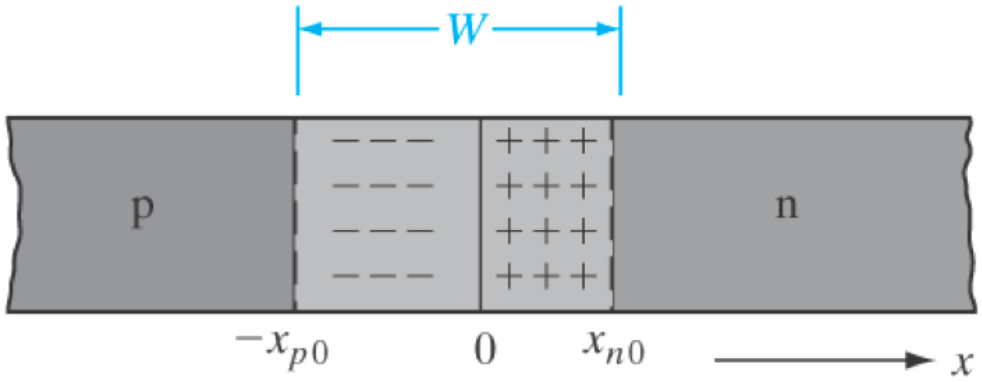


Space Charge at the Junction

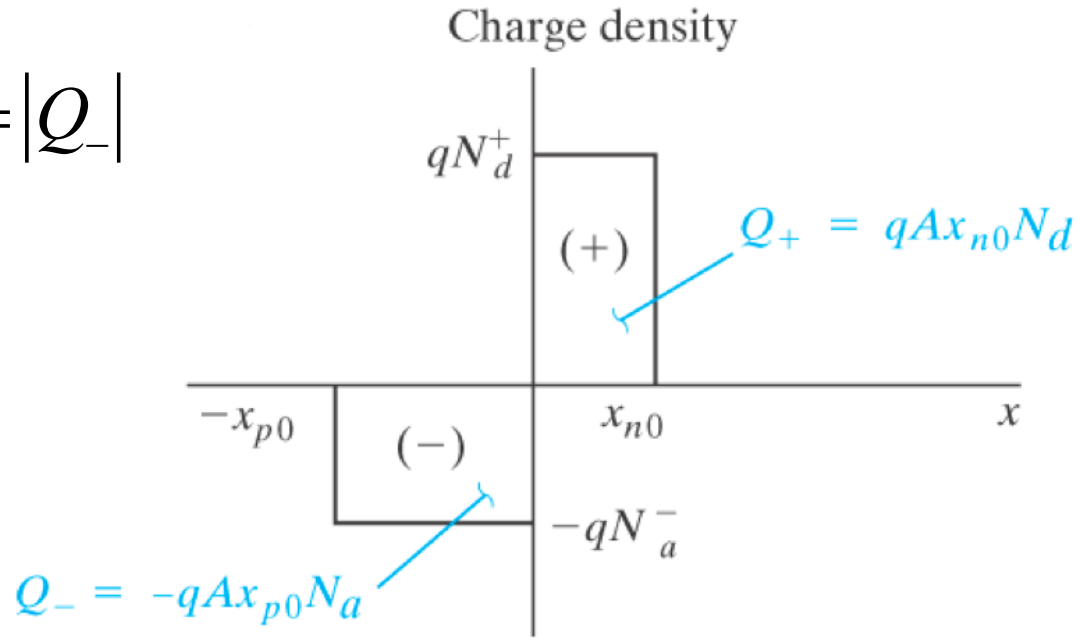
Establishing Equilibrium

- When the materials are “joined” diffusion occurs across the metallurgical junction
- Carriers that diffuse across the junction become minority carriers and recombine
- The density of free majority carrier electrons and holes becomes lower than the ionized impurity concentration near the metallurgical junction, and a field begins to form
- The field causes a drift current to form that is in the opposite direction of the diffusion current
- Equilibrium is re-established when drift exactly balances diffusion

Space Charge

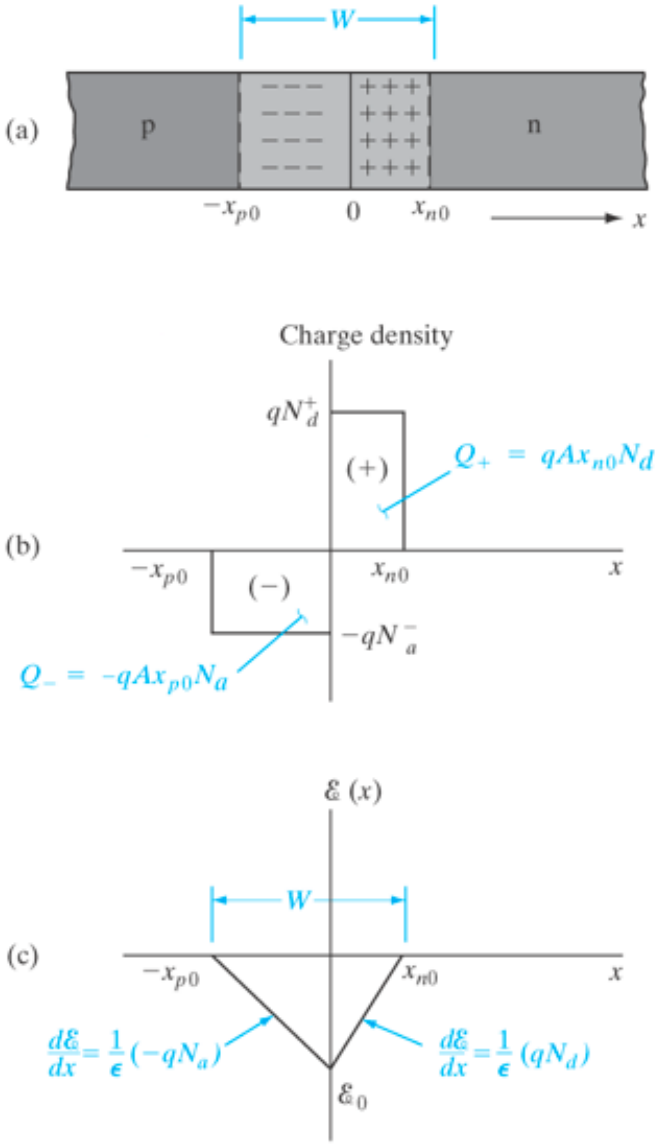


$$|Q_+| = |Q_-|$$



The Transition Region

- Within the depletion region, there are very few free carriers
- The electric field within the depletion region sweeps out any carriers which enter
- The charge density is q times the number of uncompensated donors or acceptors
- The transition width is determined by the relative concentrations
- The assumption of depletion within W and neutrality outside of W is the "***depletion approximation***"



Depletion Approximation

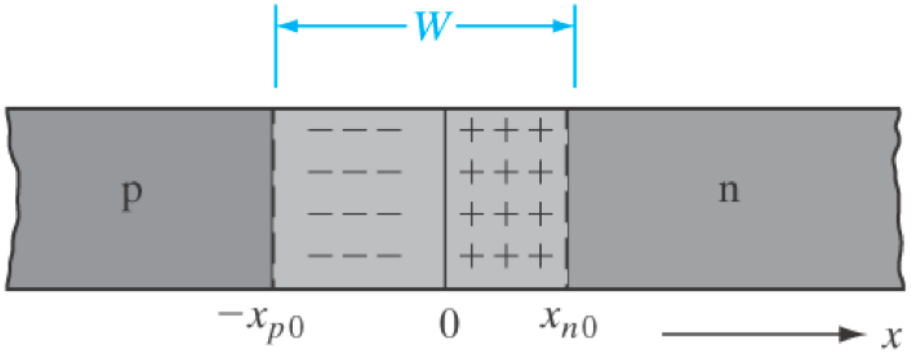
Depletion Approximation:

$$Q^+ = \left|Q^-\right|$$

$$qN_d^+ Ax_{no} = qN_a^- Ax_{po}$$

$$N_d^+ x_{no} = N_a^- x_{po}$$

$$x_{no} + x_{po} = W$$



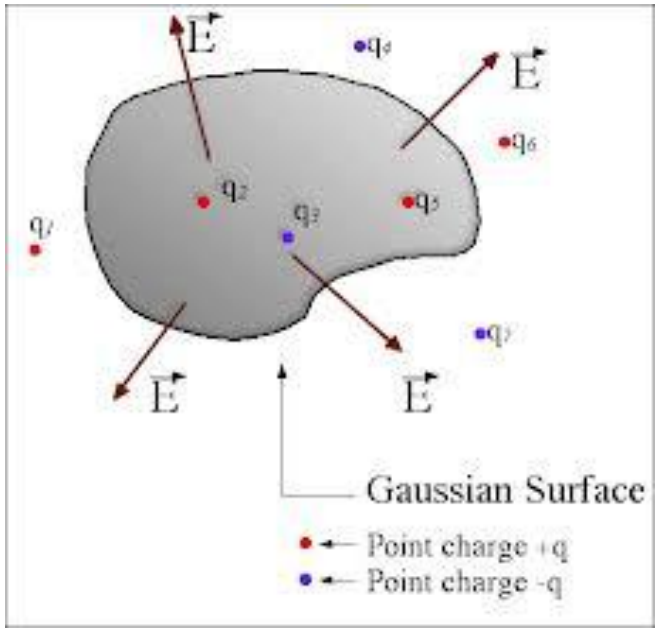
- The depletion approximation states that carriers are depleted within the depletion width “W” and that charge neutrality applies outside of the depletion region

Reminder: Gauss' Law

$$\nabla \cdot \mathbf{D} = \rho \quad \text{or} \quad \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\varepsilon}$$

ρ is the charge density with units of $\frac{Coulombs}{m^3}$

- The divergence of the electric displacement flux density is equal to the charge density
- Can be derived from Ampere's Law using the Continuity Equation

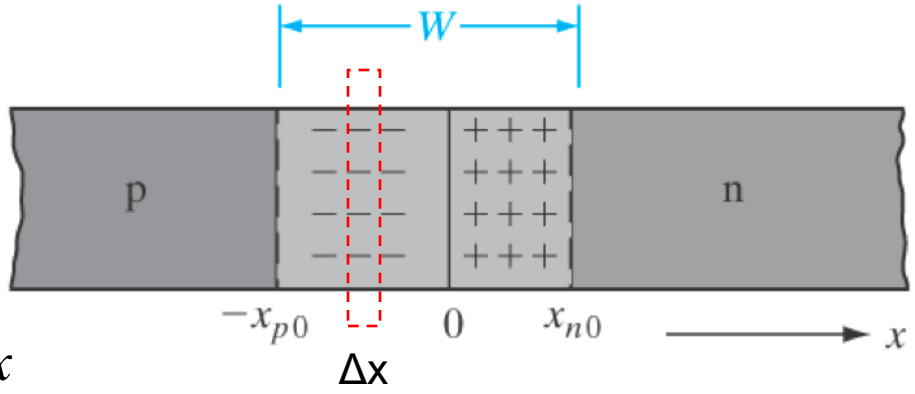


http://www.ibiblio.org/links/devmodules/Gauss/graphics/blob1.gif

Poisson's Equation

Apply Gauss's Law: $\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q}{\varepsilon}$

$$\varepsilon \oint_s \mathcal{E} \bullet dS = \sum_i q_i$$



For a small volume $\Delta x \bullet A$:

$$\varepsilon \cdot \Delta\mathcal{E} \cdot A = q\left[p + N_d^+ - n - N_a^- \right] \cdot A \cdot \Delta x$$

$$\frac{\Delta\mathcal{E}}{\Delta x} = \frac{q}{\varepsilon}\left[p + N_d^+ - n - N_a^- \right]$$

$$\lim_{x \to 0} \frac{\Delta\mathcal{E}}{\Delta x} = \frac{d\mathcal{E}(x)}{dx} = \frac{q}{\varepsilon}\left[p + N_d^+ - n - N_a^- \right]$$

Poisson's Equation:

$$\frac{d\mathcal{E}(x)}{dx} = \frac{q}{\varepsilon}\left[p + N_d^+ - n - N_a^- \right]$$

Calculation of Electric Field: n-Side

n-side of pn junction $(0 < x < x_{no})$:

$$p \approx 0,\ n \approx 0,\ N_a^- \approx 0,\ \frac{d\mathcal{E}(x)}{dx} = \frac{q}{\varepsilon}\left[N_d^+\right]$$

$$\int_{\mathcal{E}_o}^{0} d\mathcal{E}(x) = \frac{q}{\varepsilon}\left[N_d^+\right] \int_{0}^{x_{no}} dx$$

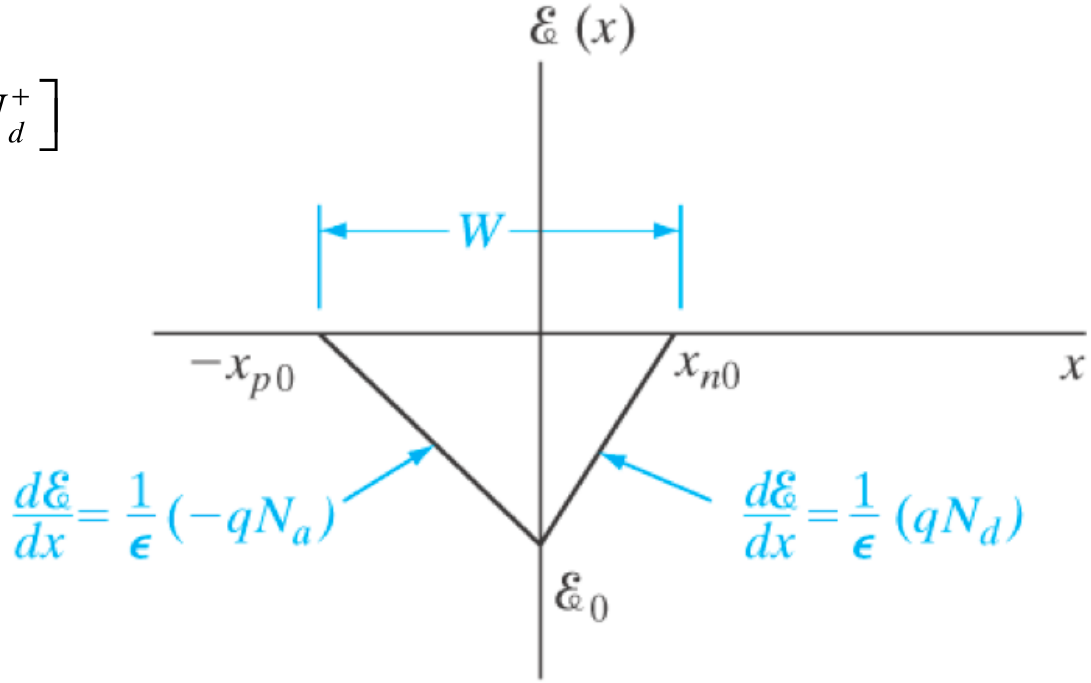
$$\left[0 - \mathcal{E}_o\right] = \frac{q}{\varepsilon}\left[N_d^+\right]\left[x_{no} - 0\right]$$

$$\mathcal{E}_o = -\frac{q}{\varepsilon}\left[N_d^+\right] x_{no}$$

To determine the value $\mathcal{E}(x)$:

$$\int_{\mathcal{E}(x)}^{0} d\mathcal{E}(x) = \frac{q}{\varepsilon}\left[N_d^+\right] \int_{x}^{x_{no}} dx$$

$$\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_d^+\right] (x_{no} - x)$$



Maximum Value of Electric Field:

$$\mathrm{E_o} = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

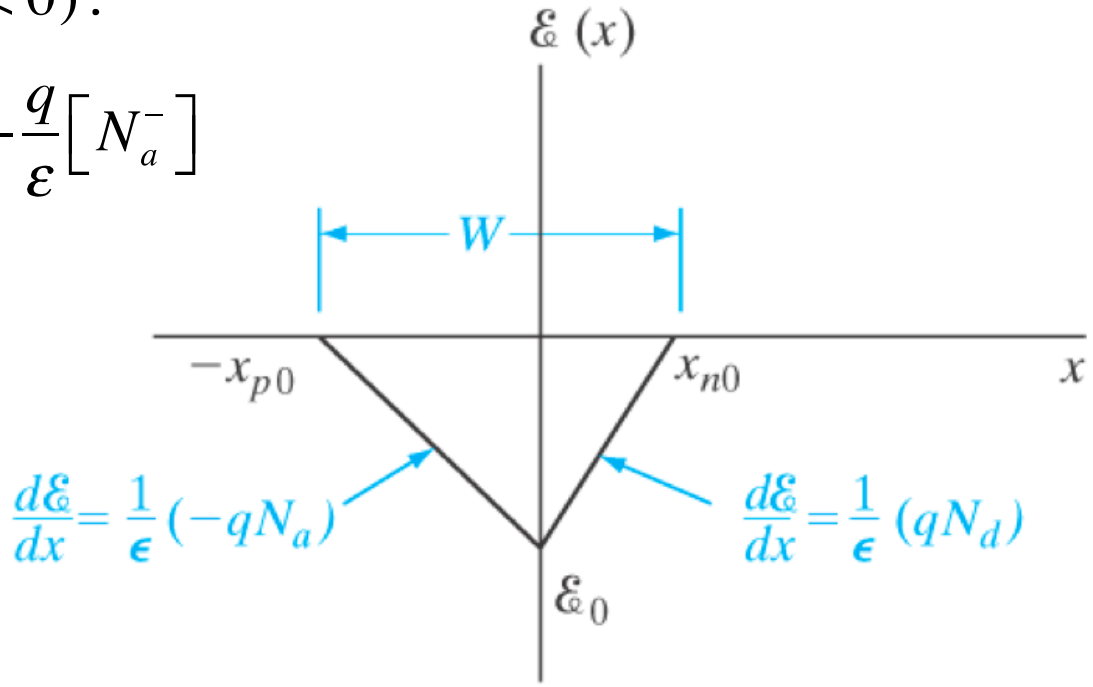
Calculation of Electric Field: p-Side

p-side of pn junction $(-x_{po} < x < 0)$:

$$p \approx 0,\ n \approx 0,\ N_d^+ \approx 0,\ \frac{d\mathcal{E}(x)}{dx} = -\frac{q}{\varepsilon}\left[N_a^-\right]$$

$$\int_0^{\mathcal{E}_o} d\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_a^-\right] \int_{-x_{po}}^{0} dx$$

$$\mathcal{E}_o = -\frac{q}{\varepsilon}\left[N_a^-\right] x_{po}$$



To Determine $\mathcal{E}(x)$:

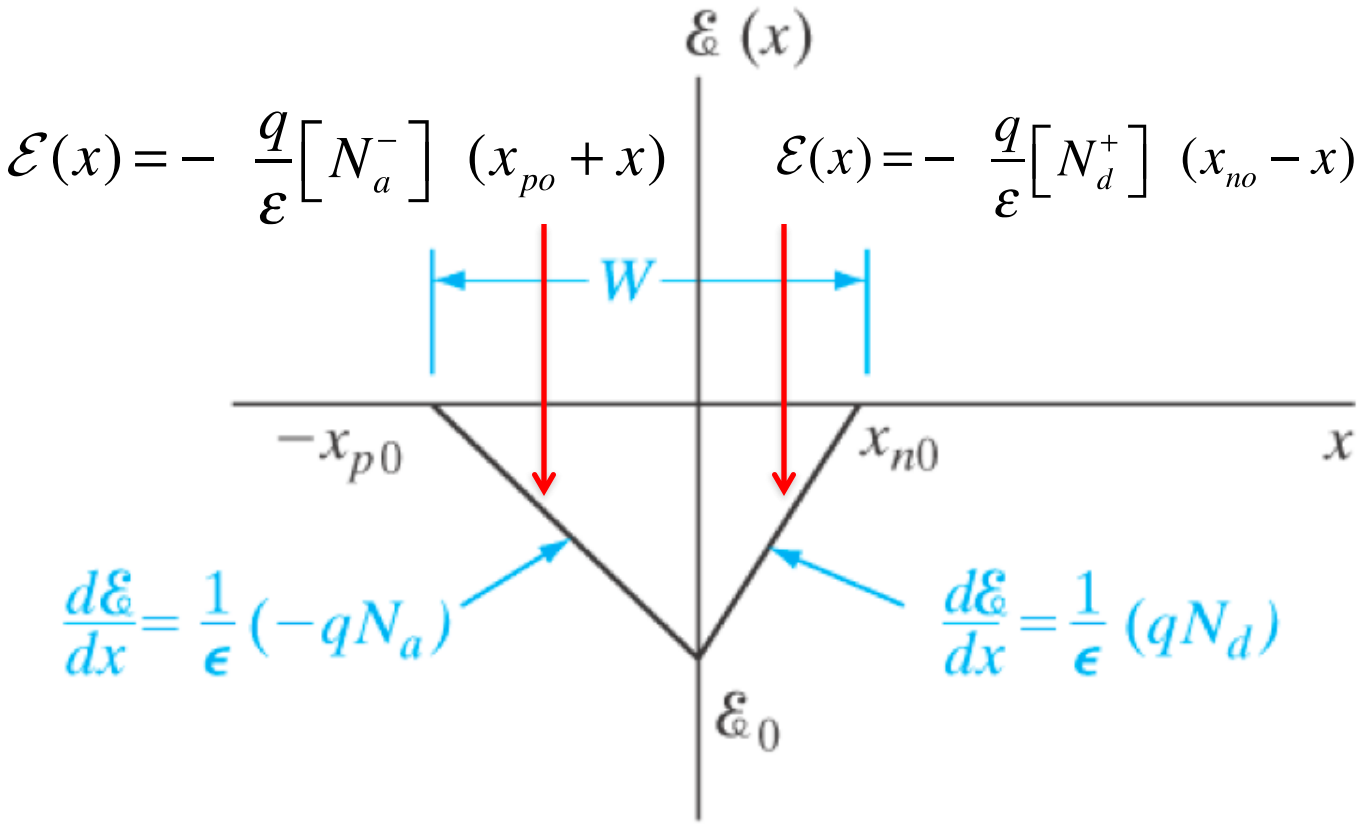
$$\int_0^{\mathcal{E}(x)} d\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_a^-\right] \int_{-x_{po}}^{x} dx$$

$$\mathcal{E}(x) = -\frac{q}{\varepsilon}\left[N_a^-\right] (x_{po} + x)$$

Maximum Value of Electric Field:

$$\mathrm{E_o} = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

Junction Electric Field: Summary



$$\mathcal{E}_o = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

From Integrating Poisson's Equation

Electric Field and Contact Potential

Maximum Value of Electric Field:

$$\mathcal{E}_o = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

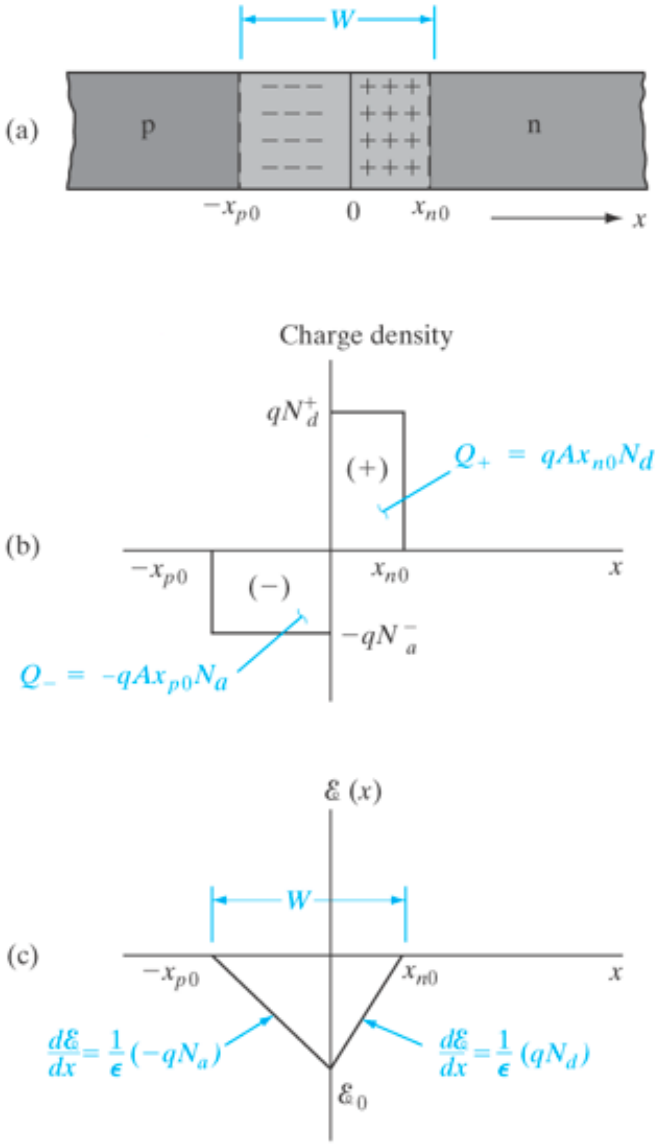
Relationship of Electric Field to Contact Potential:

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx} \text{ or } -V_o = \int_{-x_{po}}^{x_{n0}} \mathcal{E}(x)\,dx$$

Since the electric field is the negative of the potential gradient at any point.

Recall that:

$$V_o = \frac{kT}{q} \ln \frac{p_p}{p_n} = \frac{kT}{q} \ln \frac{n_n}{n_p}$$



Calculation of Depletion Width

From the previous slide:

$$V_o = -\int_{-x_{po}}^{x_{n0}} \mathcal{E}(x)\,dx$$

So, the negative of the contact potential is the area under the $\mathcal{E}$(x) versus x triangle:

$$V_o = -\frac{1}{2}\mathcal{E}_o \cdot W = \frac{1}{2}\frac{q}{\varepsilon}N_d x_{no} W = \frac{1}{2}\frac{q}{\varepsilon}N_a x_{po} W$$

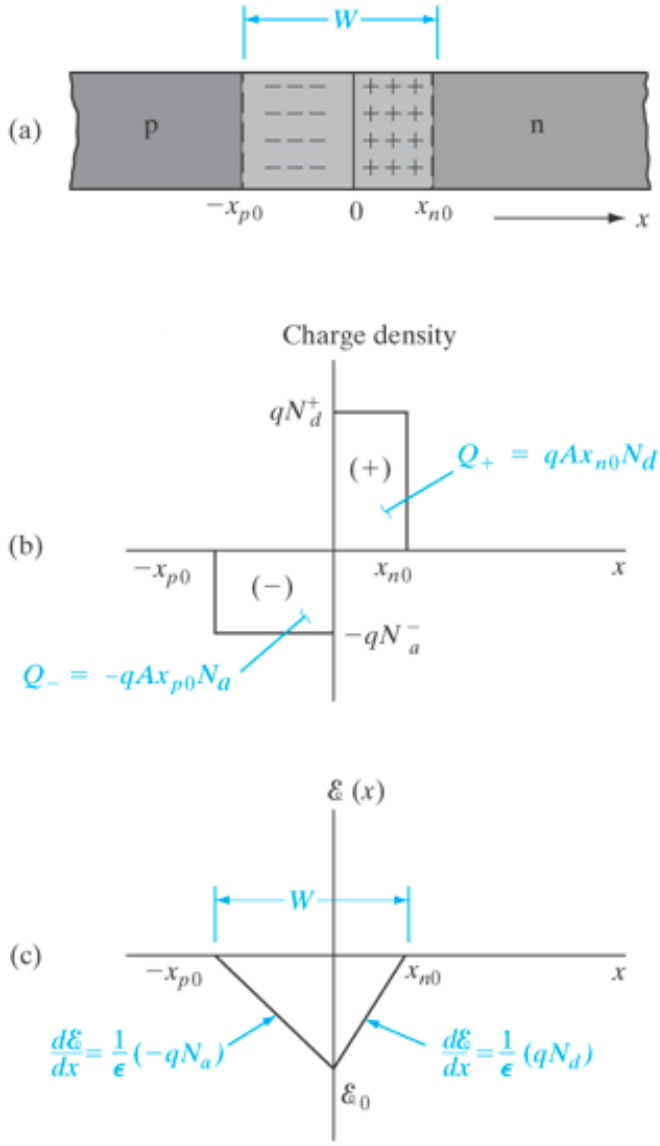
Balance of Charge Requirement: $Q_+ = |Q_-| \Leftrightarrow N_d x_{no} = N_a x_{po}$

$$x_{po} = x_{no}\frac{N_d}{N_a} \text{ and } W = x_{no} + x_{po} = x_{no}\left(1 + \frac{N_d}{N_a}\right),$$

$$\text{so } x_{no} = W\left(\frac{N_a}{N_a + N_d}\right)$$

$$\text{and } V_o = \frac{1}{2}\mathcal{E}_o W = \frac{1}{2}\frac{q}{\varepsilon}\frac{N_a N_d}{N_a + N_d}W^2$$

$$W = \left[\frac{2\varepsilon V_o}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2} = \left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$



Other Useful Relationships

$$W = \left[\frac{2\varepsilon V_o}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2} = \left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

and $V_o = \frac{kT}{q}\ln\frac{N_a N_d}{n_i^2}$ so:

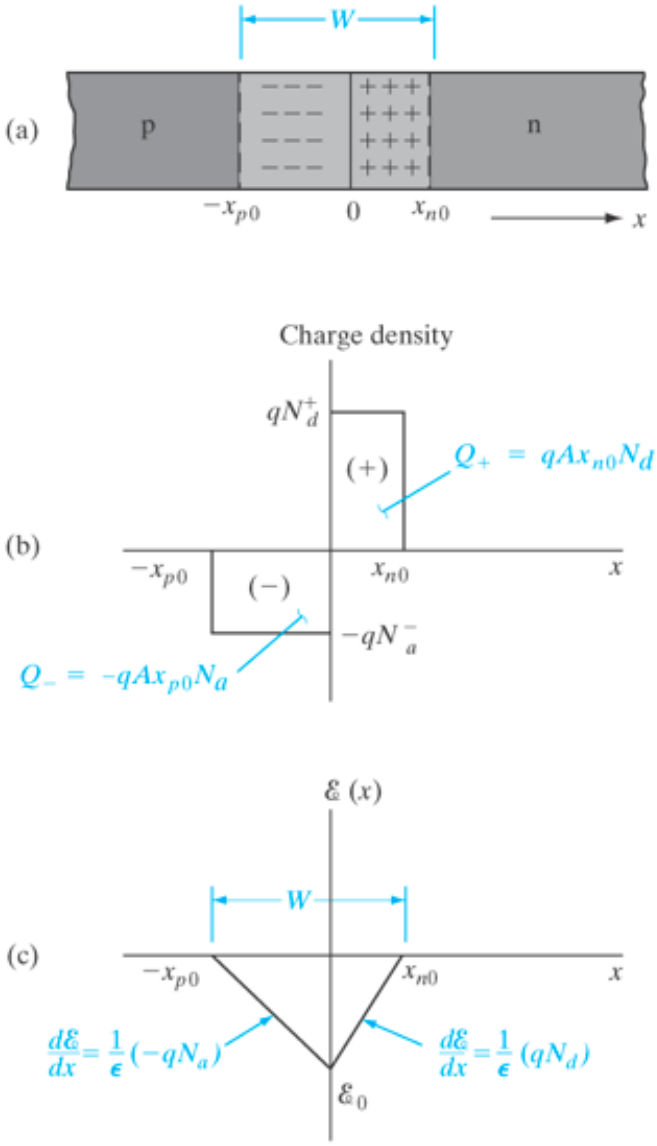
$$W = \left[\frac{2\varepsilon kT}{q^2}\left(\ln\frac{N_a N_d}{n_i^2}\right)\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$

Also:

$$x_{po} = \frac{WN_d}{N_a + N_d} = \frac{W}{1 + N_a/N_d} = \left\{\frac{2\varepsilon V_o}{q}\left[\frac{N_d}{N_a(N_a + N_d)}\right]\right\}^{1/2}$$

and

$$x_{no} = \frac{WN_a}{N_a + N_d} = \frac{W}{1 + N_d/N_a} = \left\{\frac{2\varepsilon V_o}{q}\left[\frac{N_a}{N_d(N_a + N_d)}\right]\right\}^{1/2}$$

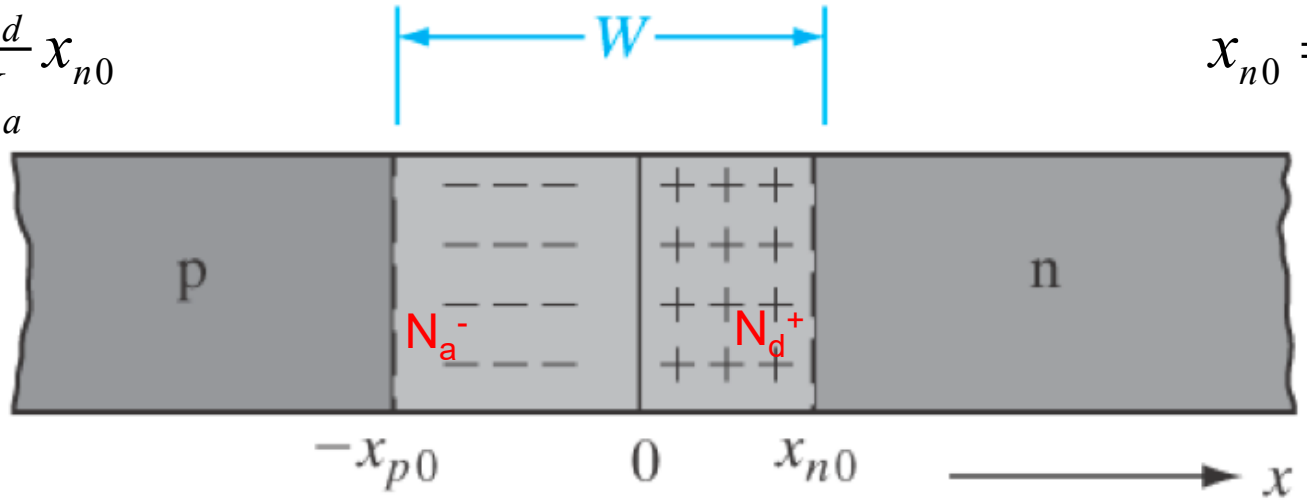


Depletion Width and Contact Potential

$$W = \left[\frac{2\varepsilon V_0}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}$$

$$x_{p0} = \frac{N_d}{N_a} x_{n0}$$

$$x_{n0} = \frac{N_a}{N_d} x_{p0}$$



$$x_{p0} = \left\{ \frac{2\varepsilon V_0}{q} \left[\frac{N_d}{N_a (N_a + N_d)} \right] \right\}^{1/2}$$

$$x_{n0} = \left\{ \frac{2\varepsilon V_0}{q} \left[\frac{N_a}{N_d (N_a + N_d)} \right] \right\}^{1/2}$$